Arnav Saraf

+1 (236) 808 5525 | arnavs1971@gmail.com | linkedin.com/in/arnav-saraf

Education

University of Waterloo Sep. 2025 – Present
BASc in Electrical Engineering *Waterloo, ON*
- Presidential Scholarship of Distinction | Varsity Athlete

Skills

PCB/CAD: Altium Designer, Eagle, AutoCAD, Fusion 360, Quartus Prime
Simulation: Ansys HFSS, Keysight ADS, LTspice, GNU Radio
Programming: Python, C++, Java, Verilog
Lab/Hardware: VNAs, oscilloscopes, DMMs, signal generators, PCB assembly

Experience

Engineering Intern Sep. 2026 – Present
MPI Morheat *Toronto, ON*
- Automated heater power calculations and quoting workflows using **Java and Python**.
- Drafted sensor and heater concepts and incorporated design revisions in **AutoCAD and Fusion 360**, following company drafting conventions.
- Soldered connectors for RTD and thermocouple sensor assemblies.

Electrical Engineering Intern Jan. 2026 – May 2026
Quantum Technology Corp. *Squamish, BC*
- Designed electrical control panels in **AutoCAD Electrical** and assembled panels for industrial gas purification and liquefaction systems, following applicable IEC, CSA, and UL requirements.
- Integrated **PLCs**, process instrumentation, motor starters, VFDs, and power-distribution equipment.
- Tested panels and PCBs using oscilloscopes and bench equipment; documented results under quality-control procedures.

Electrical Engineering Team Member Sep. 2025 – Dec. 2025
Waterloo Aerial Robotics Group *Waterloo, ON*
- Designed voltage-regulation and data-acquisition PCBs for autonomous flight hardware in **Altium Designer**, including schematic capture, component selection, and layout.
- Assembled and soldered custom PCBs; performed continuity checks and functional testing.

Projects

ADS-B Receiver
- Designed and simulated a patch antenna and coupled-line bandpass filter in **Ansys HFSS** for 1090 MHz ADS-B reception; obtained a simulated center frequency of 1090.2 MHz and return loss of 18 dB.
- Laid out the antenna and filter PCB in **Altium Designer** for manufacturing; measured prototype S-parameters with a **VNA** and achieved return loss within 8% of the simulated values.
- Developed a **GNU Radio** DSP flowgraph to demodulate and decode live ADS-B transmissions, extracting aircraft call-sign, position, and altitude

High-Frequency Doppler Radar
- Designed and manufactured a transmitter/receiver PCB in Altium Designer with impedance-controlled microstrip routing, low-noise amplification, and filtering for 2.4 GHz.
- Simulated mixer and IF processing stages in **Keysight ADS**, achieving a 1 Vpp output for Arduino-based measurement of Doppler shifts up to 1 kHz.
- Optimized patch antenna geometry and placement in **Ansys HFSS**, evaluating radiation patterns and achieving 22 dB Tx/Rx isolation at 2.4 GHz.

FPGA-Based Heart-Rate Monitor
- Implemented heartbeat detection and BPM calculation from successive beat intervals in **Verilog**, verifying functionality through simulation.
- Developed and simulated **Verilog** digital filters to reduce noise in ECG signals.
- Implemented an **SPI** acquisition module to deserialize ADC output into 12-bit parallel samples for FPGA processing.